

1. Randomisation testing in four steps

1. Assume no treatment effect

Fisher's sharp null says treatment changes nobody's outcome:

$$H_0 : Y_i(1) = Y_i(0) \text{ for every unit } i.$$

Under this assumption, outcomes stay fixed when we move the treatment labels.

2. Calculate what happened

Use the observed treatment-control difference:

$$\hat{\tau}_{\text{obs}} = \bar{Y}_{T=1} - \bar{Y}_{T=0}.$$

Could random assignment alone easily have produced a difference this large?

3. Re-run the random assignment

Reassign treatment in every way allowed by the experiment. Each time, recalculate the treatment-control difference.

These hypothetical differences show what the experiment could produce when treatment has no effect.

4. Count results at least as extreme

$$p = \frac{\# \text{ assignments at least as extreme}}{\# \text{ allowable assignments}}.$$

A small p -value means: the observed difference would be unusual if treatment truly affected nobody.

For a two-sided test, count extreme results in both directions.

2. Construct the exact p -value: count extreme assignments

Build the denominator

If exactly n_T of N units are treated, the assignment space contains

$$\binom{N}{n_T} = \frac{N!}{n_T!(N - n_T)!}$$

treatment allocations. For example,

$$\binom{7}{2} = 21,$$

so assigning two of seven villages creates 21 equally likely allocations. These 21 allocations form the denominator of the exact p -value.

Build the numerator

For every allocation, keep the outcomes fixed under H_0 , recompute $\hat{\tau}^*$, and flag whether it is at least as extreme as the observed 6.5:

$$I^* = \mathbf{1}\{|\hat{\tau}^*| \geq 6.5\}.$$

Then

$$p = \frac{\sum_{\mathbf{T}^*} I^*}{21}.$$

Preserve the design: keep the treated count fixed and respect any blocks or clusters.

Village example

Two villages receive female-headed councils. The observed difference in means is 6.5 percentage points. Under the sharp null, the 21 allocations yield:

$\hat{\tau}$	-7.5	-4	-0.5	3	6.5	10
Frequency	3	5	6	2	3	2

For a two-sided alternative, count both tails. The extreme assignments are those producing -7.5, 6.5, or 10:

$$p_{\text{two-sided}} = \frac{3 + 3 + 2}{21} = \frac{8}{21} \approx 0.38.$$

For the pre-specified upper-tail alternative, count only 6.5 and 10:

$$p_{\text{upper}} = \frac{3 + 2}{21} = \frac{5}{21} \approx 0.24.$$

Meaning: if the sharp null were true, 8 of the 21 legal assignments would look at least this extreme on a two-sided test. **Choose the tail from H_A , not whichever gives the smaller p -value.**

3. Count every assignment—or simulate many of them

Small assignment space: exact test

List every legal treatment assignment and recalculate the statistic for each one.

$$p = \frac{\# \text{ extreme assignments}}{\# \text{ all assignments}}.$$

This is an **exact randomisation test** because it checks the complete assignment space.

- ▶ Works in small samples.
- ▶ Requires no normality assumption.
- ▶ Uses the actual experimental design.

Limitation: with few assignments, only a small set of p -values is possible.

Large assignment space: simulation

If there are too many assignments to list, randomly generate a large number B of legal assignments. If r simulated assignments are at least as extreme as the observed result, use

$$\hat{p} = \frac{r + 1}{B + 1}.$$

The $+1$ includes the observed assignment and prevents a simulated p -value of exactly zero.

- ▶ More simulations give a more stable estimate.
- ▶ Every simulation must follow the original assignment rule.
- ▶ Report that the p -value was simulated.

Same idea in both cases: count the no-effect assignments that look at least as extreme as the real experiment.

4. Use the p -value for a decision—then interpret it carefully

Compare with a pre-specified significance level

Choose α before seeing the result:

$$p \leq \alpha \Rightarrow \text{reject } H_0, \quad p > \alpha \Rightarrow \text{do not reject } H_0.$$

Under H_0 , a level- α test falsely rejects with probability at most α over repeated random assignments.

Read the statement in the correct direction

The p -value is

$$P(|S^*| \geq |S_{\text{obs}}| \mid H_0, \text{ assignment design}).$$

It is *not* $P(H_0 \mid \text{data})$, the probability the finding is due to chance, or the size and importance of the effect.

A large p -value means “not unusual under H_0 ,” not “ H_0 is proven.”

Invert tests to obtain an interval

Test a grid of constant-effect hypotheses:

$$H_0(\tau^*) : Y_i(1) - Y_i(0) = \tau^* \quad \forall i.$$

For each candidate τ^* :

1. adjust outcomes to remove the hypothesised effect;
2. run the randomisation test using the original design;
3. retain values not rejected at level α .

Retained values form a $(1 - \alpha)$ randomisation-based confidence set under the constant-effect model.

Test–interval duality: a value lies outside the 95% interval exactly when its two-sided test rejects at 5%.

Report the whole result

- ▶ the observed effect estimate and test statistic;
- ▶ H_0 , H_A , tail rule, and assignment scheme;
- ▶ exact or simulated p -value, plus a confidence interval.

Rejecting the sharp null says at least one unit was affected; it does not establish a constant effect or describe heterogeneity.